

Using ABARS for Application Backup and Recovery

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- **Terminology**
- **Application Centric Disaster Recovery Support**
- **ABARS Functional Components**
- **Aggregate Backup**
- **Aggregate Recovery**
- **ICF Catalog Backup and Recovery Support**

Terminology

ABARS – Aggregate Backup and Recovery Support

- No-cost component of DFSMSHsm
- DFDSS used to logical dump data on primary DASD
 - ✓ Exits and patches can be used to customize processing
 - ✓ Can *skip* data sets that would normally cause DFDSS logical dump to fail
- DFSMSHsm routines used to back up migrated (ML1/ML2) and user tape data
- Only data-mover of its kind

Terminology

Aggregate

- Collection of data sets that are backed up and recovered as a group
- Typically grouped by application

Aggregate Group

- DFSMS construct used to define backup attributes
 - ✓ Selection Data Set name(s)
 - ✓ Output data set prefix
 - ✓ Data set serialization requirements
 - ✓ Number of concurrent copies to be created
 - ✓ Concurrent Copy or SnapShot usage requirements

Terminology

Selection Data Set (SDS)

- Sequential data set or PDS member containing the list of data sets to be backed up
 - ✓ Data set name patterns supported
 - PAY*.TEST%.**
- INCLUDE, ALLOCATE, ACCOMPANY controls
 - ✓ EXCLUDE logic supported

ISMF – Interactive Storage Management Facility

- Used to define Aggregate Group and Management Class constructs

Application Centric Disaster Recovery Support

Back up specific data sets rather than volumes

- All data to be backed up must be cataloged

Data sets anywhere in the storage hierarchy

- Primary DASD, DFSMSHsm migration (ML1, ML2), user tape
- SMS and non-SMS data supported

Synchronized set of related data set backups

- Point in time determined by specific application requirements
- At the beginning or end of a batch cycle
- At the beginning or end of transaction based processing

Application Centric Disaster Recovery Support

Backups can be synchronized with database backups

- Concurrent with image copy backups
- Provides a synchronized recovery point

Applications can be recovered in priority order

- Recover most critical applications
- Activate most critical applications
- Recover less critical applications
- Activate less critical applications

Application Centric Disaster Recovery Support

New challenges

- Defining application data requirements
 - ✓ Identification of critical data sets
 - ✓ Identification of related application feeds
- Tracking application changes
 - ✓ Maintenance (new or changed jobs)
 - ✓ Cycle-to-cycle behavior changes
 - Daily, weekly, monthly, quarterly, annual

Application Centric Disaster Recovery Support

New challenges (continued)

- Identifying exposures
 - ✓ Data set overlap between multiple applications
 - Potential data set back-leveling during recovery
 - ✓ Required data sets missing from backups
 - Common reason for recovery exercise failures
- Documenting application recovery procedures
 - ✓ Application order, dependencies, etc.
- Avoiding and resolving recovery conflicts

ABARS Functional Components

Aggregate Group definition

- Defined as an SMS construct in DFSMS SCDS
- Output data set name prefix
- Processing requirements
- Selection Data Set name(s)

Aggregate Backup

- DFSMSHsm command (ABACKUP) driven process
 - ✓ Can be submitted using batch TSO job or ISMF panel option
- Performed by DFSMSHsm secondary address space
- SMS address space must be active

ABARS Functional Components

Aggregate Recovery

- DFSMSHsm command (ARECOVER) driven process
 - ✓ Can be submitted using batch TSO job or ISMF panel option
- Performed by DFSMSHsm secondary address space
- SMS address space need not be active
 - ✓ Unless SMS allocation support required
 - SMS-only data types
 - PDSE or Extended Format data sets
 - Any VSAM in the ALLOCATE list
 - SMS directed data set placement

ABARS Functional Components

Aggregate Recovery

- **ARECOVER command required parameters**
 - ✓ **ABACKUP output control file name and volume serial**
 - **Unless:**
 - BCDS ABR record exists**
 - or**
 - ABACKUP control file name is cataloged**

ABARS Functional Components

Selection Data Set controls

- **INCLUDE / EXCLUDE**

- ✓ Names of data sets to be copied
 - Online DASD, migrated, or user tape

- **ALLOCATE / ALLOCATEEXCLUDE**

- ✓ Names of data sets to be allocated
 - Online DASD only
- ✓ Copy only the data set characteristics
- ✓ Used for user catalogs, databases, and output files
- ✓ Non-SMS managed VSAM not supported

ABARS Functional Components

Selection Data Set controls (continued)

■ ACCOMPANY / ACCOMPANYEXCLUDE

- ✓ Tape data only
- ✓ Used to catalog tape data sets
 - User tape data shipped to recovery site
- ✓ Often used to support
 - DB2/IMS image copies
 - Non-system volume dumps
- ✓ Can be used to catalog 'other' tapes stored in the vault

ABARS Functional Components

Aggregate Backup

- Captures cataloged data sets only
- Online DASD data sets backed up to 'D' file
- Migrated and user tape data sets backed up to 'O' file
- Informational text backed up to 'I' file
- Control information used to drive recovery written to 'C' file
- Can create up to 15 concurrent backup copies
- Up to 64 concurrent ABACKUPS can execute in each DFSMSHsm LPAR

ABARS Functional Components

- **ABACKUP processes application data sets serially**
 - ✓ Attempts exclusive ENQ for all data sets prior to copy function
- **Test backup of aggregates containing large files**
 - ✓ Determine execution time
 - ✓ Split into subset aggregates to reduce elapsed time
- **Utilizes large or small capacity tapes**
 - ✓ Must be balanced with RTO
- **Not restartable, must be rerun**
- **Specification of uncataloged data sets will cause ABACKUP to fail**
 - ✓ ARCBEEEXT exit can be used to bypass uncataloged data sets and other conditions that cause ABACKUP to fail

Exits

ARCBEEEXT will be called when:

- A DFDSS data set backup error occurs
- A DFSMSHsm CDS read error occurs
- A DFSMSHsm SDSP allocation error occurs
- A data set specified in the SDS is not cataloged
- A data set fails required serialization

Exits

ARCTVEXT is called to communicate with Tape Management Systems when ABARS releases tape volumes

- **ABACKUP versions rolled off**
- **ABACKUP versions deleted by EXPIREBV processing**
- **New backup version deleted when ABACKUP processing fails**
- **DFSMSrmm does not use exit**

For documentation on exits, see *DFSMSHsm Implementation and Customization Guide* and *DFSMS Installation Exits*

Patches

ABACKUP performs validation (by default) for all VSAM data sets specified in the INCLUDE list

Increases processing time

- **Can fail ABACKUP if VERIFY fails**
- **DFSMSHsm patch can be used to bypass validation: OY56460**

PATCH .ABRCB.+82 X'80' VERIFY(.ABRCB.+82 X'00')

Patches

Wait for tape drive availability

- When ABARS processing backs up or recovers an aggregate, one or more tape units are requested
- If a request for a tape unit cannot be satisfied (because all tape units are presently in use), the ABACKUP or ARECOVER processing will fail
- Patches exist for both ABACKUP and ARECOVER to wait for a tape drive to become available

ABACKUP: PATCH .ABRCB.+81 BITS(...1....)

ARECOVER: PATCH .ABRCB.+81 BITS(....1...)

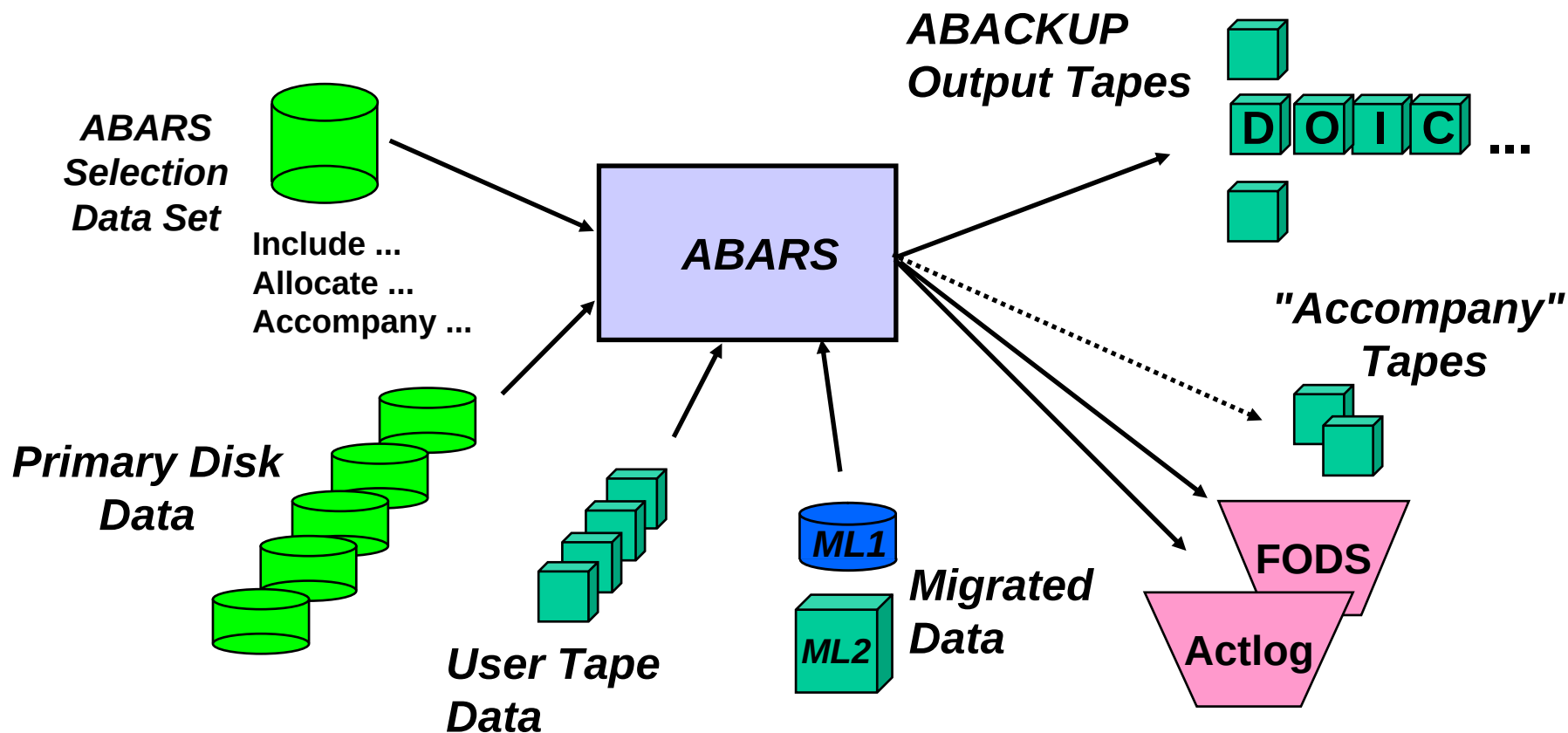
Patches

Other patches that may be of interest:

- Change the amount of time ABARS will wait for availability of an ML2 volume
- Request ABARS to provide a dump whenever specific errors occur during ABACKUP or ARECOVER processing
- Prevent ABACKUP processing from deleting existing ABACKUP output files
- Others

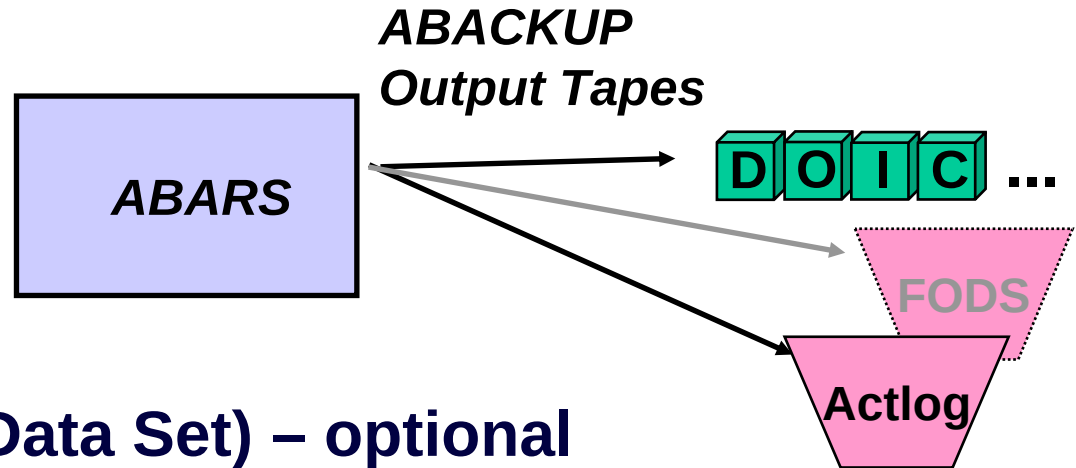
Information about patches can be found in the *DFSMShsm Implementation and Customization Guide*

ABARS ABACKUP



ABACKUP Output Files

- D file (Data)
- O file (Offline)
- I file (Instruction)
- C file (Control)
- FODS (Filter Output Data Set) – optional
- Activity Log

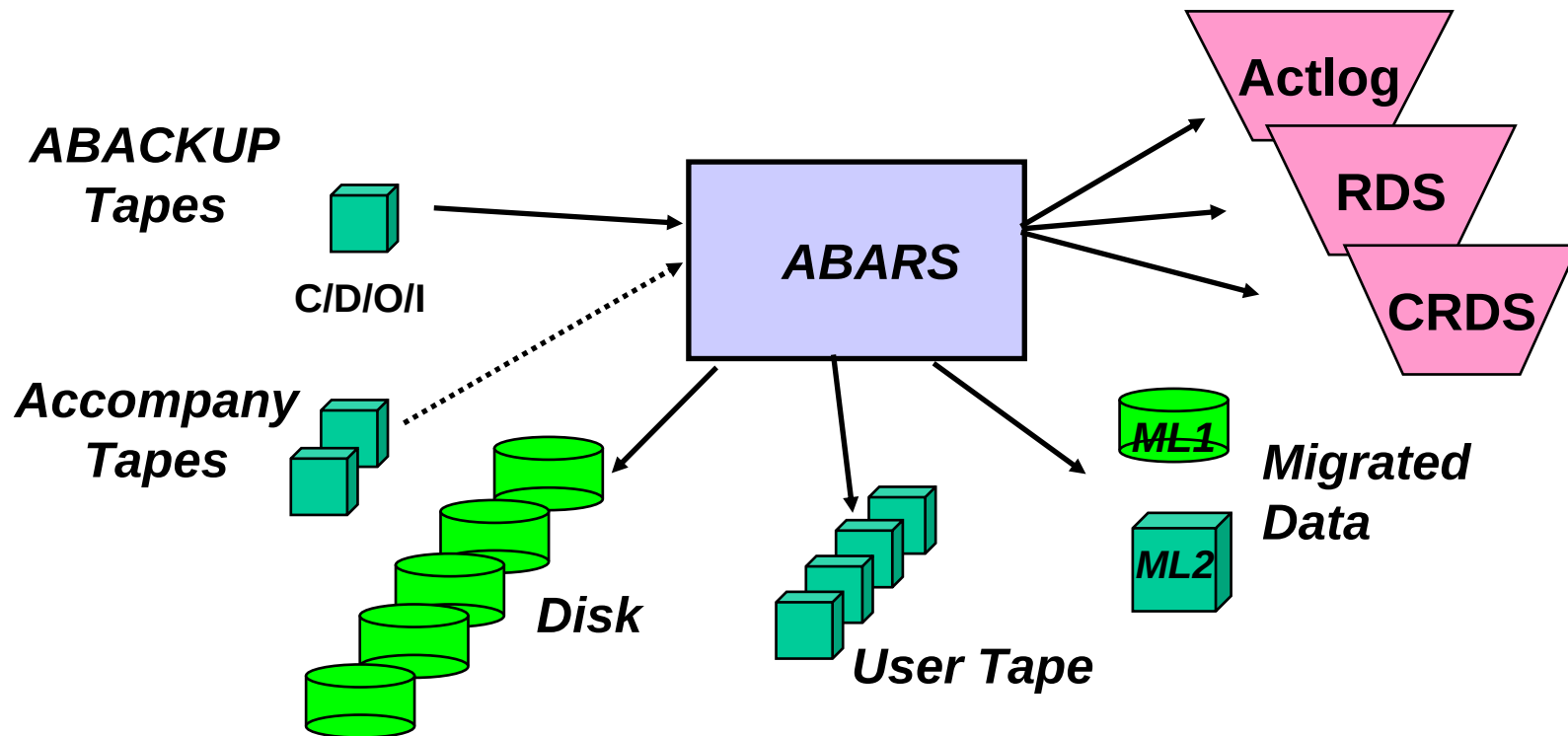


- ✓ mcv tactn.Hmcvthost.ABACKUP.agname.Dyyddd.Thhmmss
- ✓ Example:
 - HSMACT.H1.ABACKUP.TESTAG1.D06041.T164200

ABACKUP Output Files

- **D File:** outputdatasetprefix.D.CnnVnnnn
- **O File:** outputdatasetprefix.O.CnnVnnnn
- **I File:** outputdatasetprefix.I.CnnVnnnn
- **C File:** outputdatasetprefix.C.CnnVnnnn
- **STACK** feature attempts to place all output files on the same tape
 - ✓ If insufficient capacity, span to multiple tapes
 - ✓ STACK is the default

ABARS ARECOVER



ARECOVER

- Reads control and data files created by aggregate backup
- The DFSMSHsm BCDS ABR record contains ABACKUP information that can be used by ARECOVER
 - ✓ ABACKUP output volume control file name and volume serial
- If the ABACKUP control file name is cataloged, use ARECOVER with XMIT parameter
- If the ABR record does not exist and the ABACKUP control file is not cataloged, supply ARECOVER command parameters
 - ✓ ABACKUP output volume control file name and volume serial
- Several other optional parameters are available to control ARECOVER processing

Aggregate Recovery Processing:

- Recovers and catalogs all data sets in INCLUDE list
- Allocates and catalogs all data sets in ALLOCATE list
- Catalogs all data sets in ACCOMPANY list
- Recovers data to SMS or non-SMS managed DASD, DFSMSHsm migration, and user tape
- Recovers DFSMSHsm migrated data sets
 - ✓ To the level from which they were backed up
 - ✓ ALL to ML1
 - ✓ ALL to ML2
- Defines GDG bases if not present in catalog
 - ✓ During GDS recovery

Restart Environment for ARECOVER

■ Restart Data Set

- ✓ Contains list of data sets successfully recovered
 - ARECOVER restart will bypass these data sets
- ✓ Can be modified to customize ARECOVER restart processing
 - Delete a data set name – recover a data set already recovered
 - Add a data set name – skip a data set not already recovered
- ✓ Deleted as a result of successful ARECOVER processing (RC=0)

■ Conflict Resolution Data Set

- ✓ Contains list of data sets that failed recovery
- ✓ Update to specify restart actions for each data set
 - BYPASS, REPLACE, RENAMETARGET or RENAMESOURCE
 - BYPASS is default
- ✓ Deleted as a result of successful ARECOVER processing (RC=0)

Conflict Resolution Options for ARECOVER Restart

- When encountering cataloged data sets during recovery
 - ✓ **BYPASS**
 - Retain cataloged data set and skip recovery of incoming data set
 - ✓ **REPLACE**
 - Delete cataloged data set and recover incoming data set
 - ✓ **RENAMETARGET**
 - Rename cataloged data set and recover incoming data set
 - ✓ **RENAMESOURCE**
 - Retain cataloged data set and rename incoming data set

Online DASD data placement

- Controlled by SMS ACS routines at recovery site
- SMS data sets
 - ✓ Recovered to appropriate Storage Groups
- Non-SMS data sets
 - ✓ Recovered to source volume if present
 - ✓ Recovered to DFSMSHsm ARPOOL if defined
 - Volume utilization balancing parameter (PERCENTUTILIZED) available

Migrated data placement

- Default level is same as source data set level
 - ✓ ML1 to ML1
 - ✓ ML2 to ML2
- Can force all to a specific level
 - ✓ All to ML1
 - ✓ All to ML2

User tape data placement

- ✓ Recovered to scratch tapes
 - Same or different device types
 - Single or multivolume
 - File sequencing preserved

ARECOVER Output

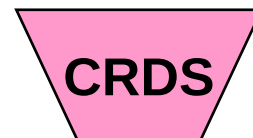
ARECOVER Activity Log



Restart Data Set



Conflict Resolution Data Set



ARECOVER Output

Activity Log data set (ACTLOG)

mcvtactn.Hmcvthost.ARECOVER.agname.Dyyddd.Thhmmss
HSMACT.H1.ARECOVER.TESTAG1.D06051.T081200

Restart Data Set (RDS)

authid.agname.RESTART.Dyyddd.Thhmmss
HSM.TESTAG1.RESTART.D06041.T164200

Conflict Resolution Data Set (CRDS)

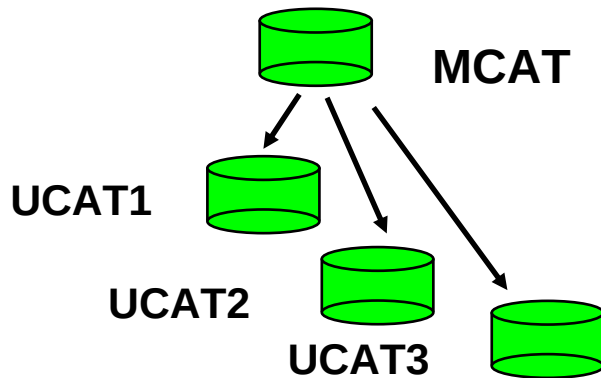
authid.agname.CONFLICT.Dyyddd.Thhmmss
HSM.TESTAG1.CONFLICT.D06041.T164200

Add User Catalogs to ABACKUP ALLOCATE list

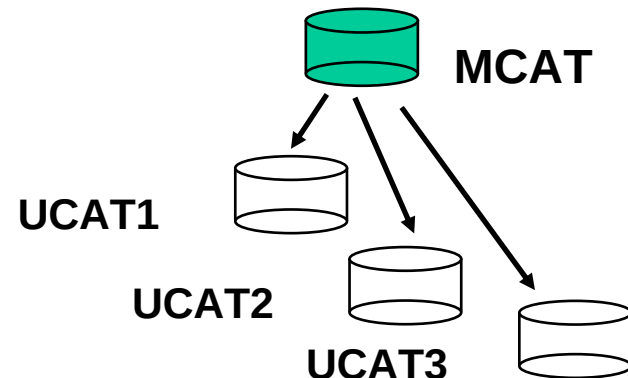
- ABACKUP will capture catalog allocation attributes and related ALIAS information
- ARECOVER will allocate catalogs and connect all related ALIASes
- Catalog contents can be restored after ARECOVER processing
 - ✓ Batch catalog recovery jobs
 - ✓ Recover catalogs first, then applications
- All data sets required during IPL must be cataloged in restored base system

Catalog Backup and Recovery

Primary System



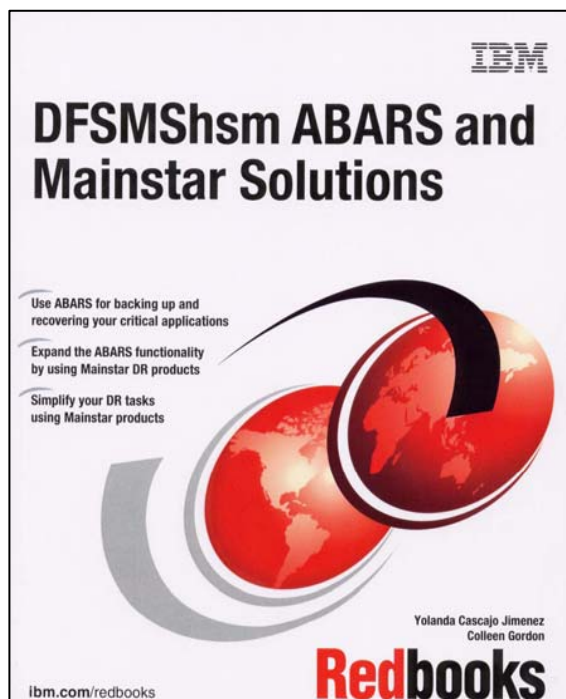
Recovery System



"ALLOCATE
(
UCAT1
UCAT2
UCAT3
)"

→ **ABARS** → ARECOVER

DFSMShsm ABARS and Mainstar Solutions



Download it at the
IBM Redbooks website
www.ibm.com/Redbooks
Look for SG24-6425

Thank you for attending SHARE

This is the end of
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Using ABARS for Application Backup and Recovery

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